34) Find a common domain for the variables x, y and z for which the statement $\forall x \forall y((x \neq y) \rightarrow \forall z((z=x) \vee (z=y)))$ is true and another domain for which it is false.

$\boxed{\text{Soln:}}$ The statement $\forall x \forall y((x \neq y) \rightarrow \forall z((z=x) \vee (z=y)))$ is true if and only if the common domain consists of atmost 2 elements.

$\boxed{\text{Proof:}}$ Let, the common domain be of atmost 2 elements.

$\boxed{\text{Case 1:}}$ Wlog, let common domain be a_1.

$\forall x \forall y((x \neq y) \rightarrow \forall z((z=x) \vee (z=y)))$ can be rewritten as

$((a_1 \neq a_1) \rightarrow ((a_1 = a_1) \vee (a_1 = a_1)))$. Since $a_1 \neq a_1$ is False, so, by defn. of conditional, $((a_1 \neq a_1) \rightarrow ((a_1 = a_1) \vee (a_1 = a_1)))$ is True.

$\boxed{\text{Case 2:}}$ Wlog, let common domain be a_1 and a_2

$\forall x \forall y((x \neq y) \rightarrow \forall z((z=x) \vee (z=y)))$ can be rewritten as

$((a_1 \neq a_1) \rightarrow ((a_1 = a_1) \vee (a_1 = a_1)) \wedge ((a_2 = a_1) \vee (a_2 = a_1))) \wedge$

$((a_1 \neq a_2) \rightarrow ((a_1 = a_1) \vee (a_1 = a_2)) \wedge ((a_2 = a_1) \vee (a_2 = a_2))) \wedge$

$((a_2 \neq a_1) \rightarrow ((a_1 = a_2) \vee (a_1 = a_1)) \wedge ((a_2 = a_2) \vee (a_2 = a_1))) \wedge$

$((a_2 \neq a_2) \rightarrow ((a_1 = a_2) \vee (a_1 = a_2)) \wedge ((a_2 = a_2) \vee (a_2 = a_2)))$

$(a_1 \neq a_1)$ is False. By defn. of conditional, $((a_1 \neq a_1) \rightarrow ((a_1 = a_1) \vee (a_1 = a_1)) \wedge ((a_2 = a_1) \vee (a_2 = a_1)))$ is True.

$(a_2 \neq a_2)$ is False. By defn. of conditional, $((a_2 \neq a_2) \rightarrow ((a_1 = a_2) \vee (a_1 = a_2)) \wedge ((a_2 = a_2) \vee (a_2 = a_2)))$ is True.

$(a_1 = a_1)$ is True. By defn. of OR, $(a_1 = a_1) \vee (a_1 = a_2)$ is True and $(a_1 = a_2) \vee (a_1 = a_1)$ is True.

$(a_2 = a_2)$ is True. By defn. of OR, $(a_2 = a_1) \vee (a_2 = a_2)$ is True and $(a_2 = a_2) \vee (a_2 = a_1)$ is True.